

Special Problem Defense

MitoChime

A Machine Learning Pipeline for Detecting PCR-Induced
Chimeric Reads in Mitochondrial Illumina Data

Duranne Duran · Yvonne Lin · Daniella Pailden

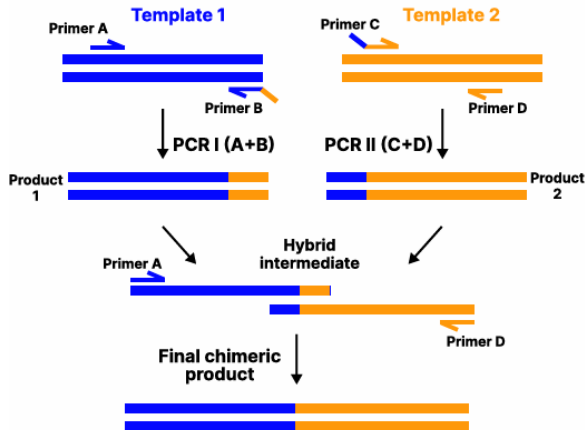
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Division of Physical Sciences and Mathematics

Four simulated *Sardinella lemuru* mtDNA short reads

Read 1	TCAGCCTTTACCAAGGCCCCACCCTTGCGGCCCTCGCCTGCTTAATTTTACTTTCTTA...
Read 2	TAAGTAAGAACTAGTAGGATCGAGGCCTTCCCGTCTAGATAGGCGTTGGCTTAATTGCA...
Read 3	ATAGGGCAGAAGGCCTAGCATGTTAATGGTAAATAGGAAAACCATTAGAGACGTAAATG...
Read 4	GAGGTGGCGAAAGTCAGCGTGGTTCCCAGTCCAAGGCTCATCAGCAAGACGGTTAGGCT...

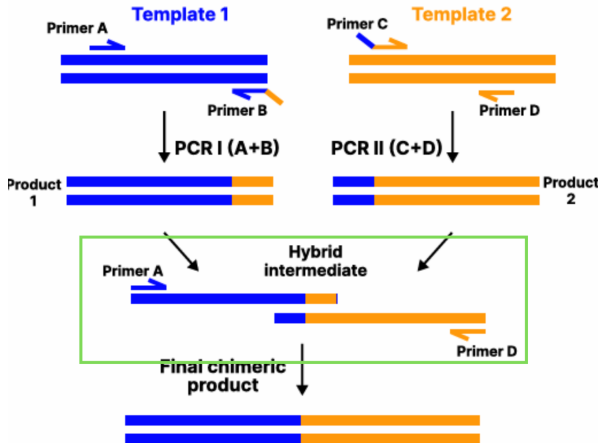
Question: Out of these short reads, which ones are chimeric?

The Problem



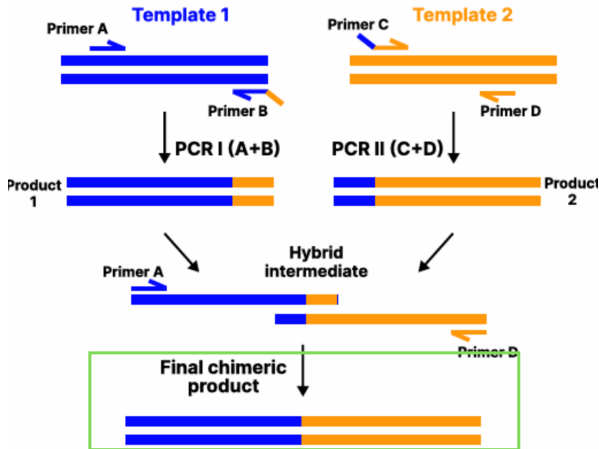
Schematic illustration of PCR-induced chimera formation through incomplete extension and template switching.

The Problem



Schematic illustration of PCR-induced chimera formation through incomplete extension and template switching.

The Problem



Schematic illustration of PCR-induced chimera formation through incomplete extension and template switching.

Implications

Assembly quality

- ▶ Fragmented or failed mitochondrial assemblies
- ▶ Reduced reliability of mitochondrial genomics workflows

Research impact

- ▶ Limited downstream analyses in fisheries, population genetics, and evolution
- ▶ Wasted sequencing effort, computing time, and researcher labor

Existing approaches

Tool / Approach	Main focus	Gap for this study
UCHIME / UCHIME2	Amplicon chimeras	Reference/abundance dependent
DADA2	ASV-level filtering	Not read-level mtDNA filtering
CATCh	16S rRNA ensemble ML	Not for mtDNA; inconsistent
ChimPipe	RNA-seq fusion/chimera detection	Not PCR-induced mtDNA artifacts
DL-based tools	DNA sequence motif prediction	Few to no tools for chimera detection
MitoChime	Read-level mtDNA chimeras	Pre-assembly ML/DL filtering

Most existing tools target amplicon, ASV, or RNA-seq chimeras; MitoChime targets PCR-induced mitochondrial short-read artifacts.

Can machine learning detect PCR-induced chimeric reads
before mitochondrial genome assembly?

Input

Simulated clean and chimeric paired-end
Illumina reads from *S. lemuru*

Output

A modular pipeline that predicts and filters
suspected chimeric reads

Scope and limitations

Included

- ▶ *Sardinella lemuru*
- ▶ mitochondrial Illumina reads
- ▶ PCR-induced chimeras
- ▶ simulated labelled datasets

Excluded

- ▶ long-read sequencing
- ▶ NUMTs
- ▶ naturally occurring chimeras
- ▶ multi-species model training

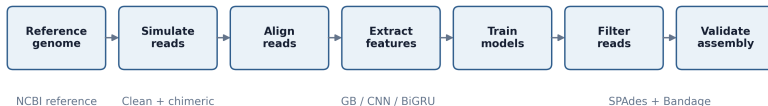
The study prioritized a focused and controlled evaluation setting.

Methodology

From simulated reads to model-based filtering

MitoChime workflow

MitoChime workflow



Overview

The workflow generated labelled reads, extracted read-level evidence, trained models, filtered suspected chimeras, and evaluated downstream assembly behavior.

Data construction

Clean reads

Simulated from the *S. lemuru* mitochondrial reference genome.

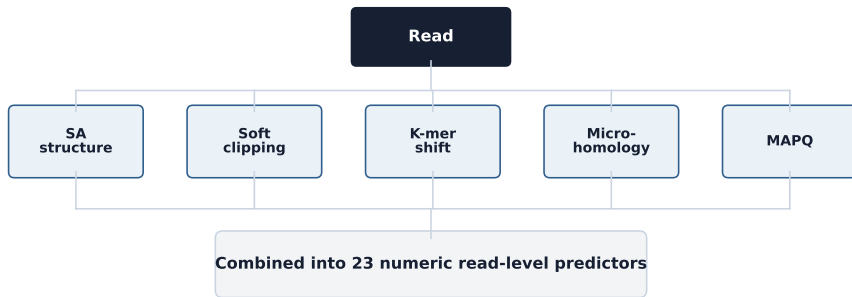
Chimeric reads

Generated by joining non-adjacent mitochondrial segments to mimic PCR template switching.

Labels: **clean** = 0 and **chimeric** = 1

Feature extraction

Read-level feature groups



Feature representation

Each read was converted into a set of 23 numeric predictors describing alignment structure, clipping behavior, sequence composition, microhomology, and mapping quality.

Dataset representation

The labelled reads were prepared in two dataset formats.

read_id	label	seq
NC_039553.1_10001_10492_5:0:0_8:0:0_b77/1	0	GTAAGGCATAGTAAGGCAATGGTGAGGCTGTGTGTGATTGCA
NC_039553.1_10001_10492_5:0:0_8:0:0_b77/2	0	GAGTGAGCCGAATAGGGAATTAGTCCAACCTAAGACTCCTGA
NC_039553.1_10003_10548_5:0:0_8:0:0_f4d/1	0	GTGAGCCGAAAAGGTAATTAGTCCAACCTACGACTCCTGATT
NC_039553.1_10003_10548_5:0:0_8:0:0_f4d/2	0	TGGCTATATAGTGTTGGTGAGGGCTCAGGCCGTTTCCGTTTC
NC_039553.1_10005_10512_6:0:0_6:0:0_c35/1	0	CCGTTGTCCAGTTTAGCCAGGTAAGGCTTAGTAAGGCAATGC
NC_039553.1_10005_10512_6:0:0_6:0:0_c35/2	0	GAGCCGCATAGGGCAATAGTCCAACCTAAGACTCGTGATTTC
NC_039553.1_10008_10523_3:0:0_10:0:0_914/1	0	TCAGCCCCTTTCCGTTGACCAGTTTAGCCAGGTAAGGCTTAC
NC_039553.1_10008_10523_3:0:0_10:0:0_914/2	0	CCGAATAGGGACATAGTCCAACCTAAGACTCCTGATTTCCTGC
NC_039553.1_10012_10513_12:0:0_5:0:0_1340/1	0	TCCGTTGTGCAGTTTAGCCAGGTAAGGCTTAGTAAGGCAATG
NC_039553.1_10012_10513_12:0:0_5:0:0_1340/2	0	ATAGTGAATTAGTCCAAGTTAAGACTCCTGATTTCGGCTCAGA
NC_039553.1_10012_10591_3:0:0_5:0:0_2571/1	0	AGCCAGCATGTAAGAACCACAGCGGGGAGGATCAGGGATC
NC_039553.1_10012_10591_3:0:0_5:0:0_2571/2	0	ATATGGAATTAGTCCAACCTAAGACTCCTGATTTCGGCTCAGA
NC_039553.1_10015_10459_2:0:0_8:0:0_253e/1	0	GTGATTGCAGCAGTTCAAAGTGACTTTTATAGCGCGAAATCAG
NC_039553.1_10015_10459_2:0:0_8:0:0_253e/2	0	GGGAATTAGTCCAACCTAAGAGTCTGATTTCGGCTCAGATG.
NC_039553.1_10015_10585_10:0:0_11:0:0_d4c/1	0	GGGAATTAGACGAACATAATCTCCTGAATTCGGCTCAGATG/
NC_039553.1_10015_10585_10:0:0_11:0:0_d4c/2	0	CATGTAAGAACCCACAGCGGGGAGGCTAAGGGATCCCTGGC

read_id	label	read_len	mean_bp	ref_name	ref_str	strand	mapq	cigar	has_si	sa_count	num_seg	sa_diff	config	sa_min_delta_pos	sa_max_delta_pos
NC_039553.1_3_540_8:0:0_6:0:0_a12	0	100	13.000	NC_039553.1	3	0	60	150M	0	0	1	0		0	0
NC_039553.1_4_430_13:0:0_11:0:0_243d	0	150	13.000	NC_039553.1	4	0	60	150M	0	0	1	0		0	0
NC_039553.1_5_405_8:0:0_11:0:0_1440	0	150	13.000	NC_039553.1	5	0	60	150M	0	0	1	0		0	0
NC_039553.1_6_523_8:0:0_9:0:0_82u	0	150	13.000	NC_039553.1	6	0	60	150M	0	0	1	0		0	0
NC_039553.1_9_574_7:0:0_7:0:0_181b	0	150	13.000	NC_039553.1	9	0	60	150M	0	0	1	0		0	0
NC_039553.1_10_501_9:0:0_8:0:0_256b	0	150	13.000	NC_039553.1	10	0	60	150M	0	0	1	0		0	0
NC_039553.1_11_509_8:0:0_11:0:0_a19	0	150	13.000	NC_039553.1	11	0	60	150M	0	0	1	0		0	0
NC_039553.1_12_427_9:0:0_9:0:0_157	0	150	13.000	NC_039553.1	12	0	60	150M	0	0	1	0		0	0
NC_039553.1_12_489_8:0:0_9:0:0_86f	0	150	13.000	NC_039553.1	12	0	60	150M	0	0	1	0		0	0
NC_039553.1_12_593_13:0:0_13:0:0_2375	0	150	13.000	NC_039553.1	12	0	24	150M	0	0	1	0		0	0
NC_039553.1_14_515_7:0:0_7:0:0_20a4	0	150	13.000	NC_039553.1	14	0	60	150M	0	0	1	0		0	0
NC_039553.1_15_525_9:0:0_9:0:0_1e76	0	150	13.000	NC_039553.1	15	0	60	150M	0	0	1	0		0	0
NC_039553.1_17_520_4:0:0_8:0:0_25c	0	150	13.000	NC_039553.1	17	0	60	148MM	0	0	1	0		0	0
NC_039553.1_18_522_8:0:0_9:0:0_c84	0	150	13.000	NC_039553.1	18	0	60	150M	0	0	1	0		0	0
NC_039553.1_18_593_8:0:0_9:0:0_sca	0	150	13.000	NC_039553.1	18	0	60	150M	0	0	1	0		0	0
NC_039553.1_18_482_8:0:0_7:0:0_1e4b	0	150	13.000	NC_039553.1	18	0	60	150M	0	0	1	0		0	0
NC_039553.1_19_516_8:0:0_8:0:0_c1a	0	150	13.000	NC_039553.1	19	0	60	150M	0	0	1	0		0	0
NC_039553.1_20_525_9:0:0_10:0:0_1b37	0	150	13.000	NC_039553.1	20	0	60	150M	0	0	1	0		0	0
NC_039553.1_21_560_10:0:0_9:0:0_189b	0	150	13.000	NC_039553.1	21	0	60	150M	0	0	1	0		0	0
NC_039553.1_23_443_2:0:0_8:0:0_1009	0	150	13.000	NC_039553.1	23	0	60	150M	0	0	1	0		0	0
NC_039553.1_25_571_7:0:0_8:0:0_1932	0	150	13.000	NC_039553.1	25	0	60	150M	0	0	1	0		0	0
NC_039553.1_28_536_8:0:0_8:0:0_182	0	150	13.000	NC_039553.1	28	0	60	150M	0	0	1	0		0	0
NC_039553.1_32_487_8:0:0_8:0:0_2005	0	150	13.000	NC_039553.1	32	0	60	150M	0	0	1	0		0	0
NC_039553.1_34_520_8:0:0_9:0:0_691	0	150	13.000	NC_039553.1	34	0	60	150M	0	0	1	0		0	0
NC_039553.1_34_452_2:0:0_16:0:0_495	0	150	13.000	NC_039553.1	34	0	60	150M	0	0	1	0		0	0
NC_039553.1_36_574_7:0:0_8:0:0_993c	0	150	13.000	NC_039553.1	36	0	60	150M	0	0	1	0		0	0

Feature matrix

For machine learning models

Sequence dataset

For sequence-based deep learning models

Models evaluated

Gradient Boosting

Feature-based model using engineered alignment and sequence-derived predictors.

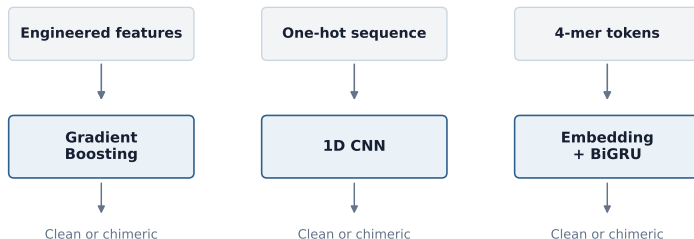
1D CNN

Sequence-based model using one-hot encoded nucleotide reads.

BiGRU

Sequence-based model using ordered k-mer tokens.

Three model representations



Evaluation strategy

Read-level evaluation

- ▶ Accuracy
- ▶ Precision
- ▶ Recall
- ▶ F1-score
- ▶ ROC-AUC

Assembly-level validation

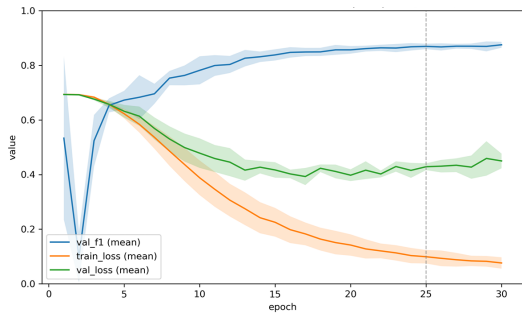
- ▶ Read retention
- ▶ Number of nodes
- ▶ Assembly graph structure
- ▶ Contiguity after filtering

The study evaluated both prediction quality and downstream utility.

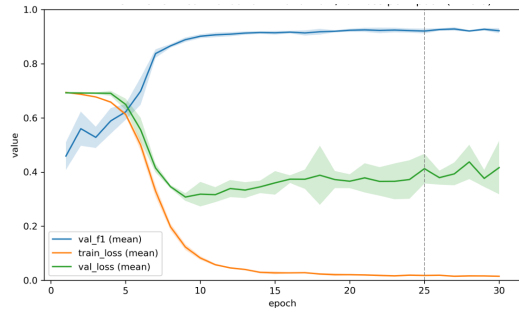
Results and Discussion

Classification performance and assembly behavior

Training Behavior of CNN and BiGRU

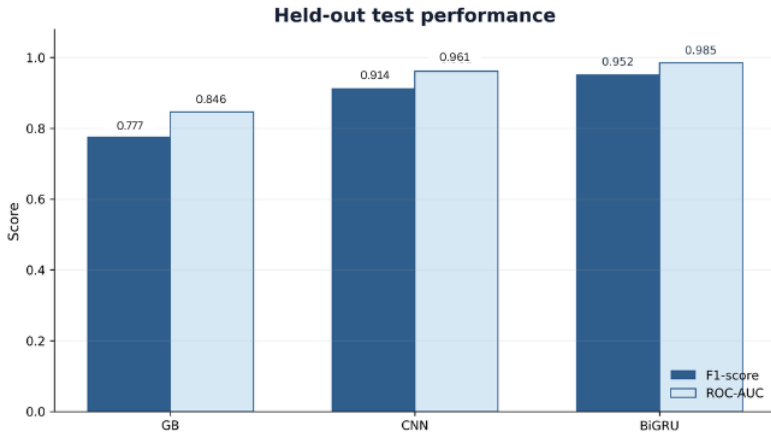


CNN



BiGRU

Classification performance

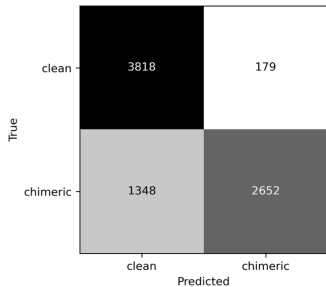


Classification performance

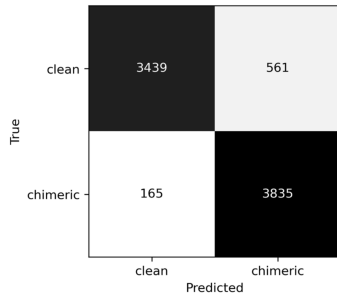
Model	F1-score	ROC-AUC	Runtime
Gradient Boosting	0.7765	0.8459	0.50 min
1D CNN	0.9135	0.9607	1.39 min
BiGRU	0.9523	0.9849	5.51 min

- ▶ **Gradient Boosting** — Conservative filtering; higher read retention
- ▶ **CNN** — Sensitive model; more aggressive filtering
- ▶ **BiGRU** — Highest F1 score; best read-level performance

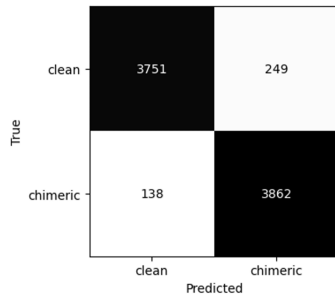
Error patterns



Gradient Boosting



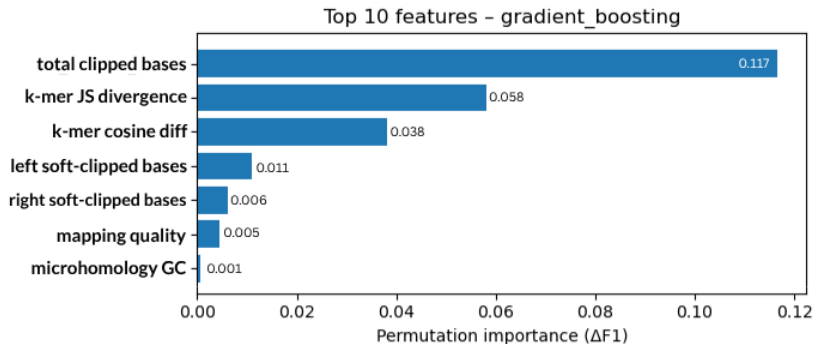
CNN



BiGRU

These classification results were reviewed with PGC Visayas and were considered suitable for the intended mitochondrial assembly workflow.

Feature importance



Most informative predictors: **total clipped bases**, **k-mer Jensen–Shannon divergence**, **k-mer cosine difference**, and **left soft clipping**.

Feature Family Importance

Family	Importance Sum	Importance Mean	nFeatures
Clipping	0.133830	0.033458	4
K-mer jump	0.096264	0.048132	2
Other	0.003676	0.001225	3
Microhomology	0.000347	0.000174	2
SA structure	-0.000859	-0.000066	13

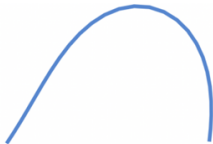
Assembly validation: high coverage



(a) Unfiltered



(b) GB



(c) CNN



(d) BiGRU

20K reads, 50% chimeras

- ▶ Unfiltered assembly showed severe graph complexity.
- ▶ Filtering simplified the graph.
- ▶ CNN produced the simplest graph.
- ▶ GB retained more reads.

Assembly validation: low coverage

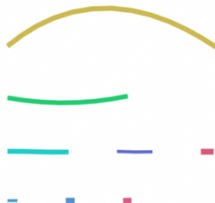


(a) Unfiltered

(b) GB



(c) CNN



(d) BiGRU

3,200 reads, 50% chimeras

At low read depth, aggressive filtering may remove too much read support and cause fragmentation.

Practical implication

The safest model depends on coverage and chimera proportion.

Recommendations: Practical Use

- ▶ Use **Gradient Boosting** when read depth is low or interpretability is important.
- ▶ Use **BiGRU** when read-level classification performance is the priority.
- ▶ **CNN** may be useful in high-coverage and high-chimera conditions.
- ▶ Deep learning models performed better, but their black-box nature makes interpretability a tradeoff.

Recommendations: Future Work

- ▶ Test on real mitochondrial datasets.
- ▶ Evaluate realistic class imbalance.
- ▶ Add varied sequencing-error profiles.
- ▶ Explore hybrid feature-sequence models.

Summary

- ① PCR-induced chimeras can disrupt mitochondrial genome assembly.
- ② Existing tools do not directly address this read-level setting.
- ③ MitoChime detects suspected chimeric reads before assembly.
- ④ BiGRU achieved the strongest classification performance.
- ⑤ Model choice should consider both accuracy and read retention.

MitoChime

A Machine Learning Pipeline for Detecting PCR-Induced
Chimeric Reads in Mitochondrial Illumina Data

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Thank you.

Appendix

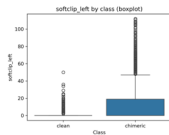
Supplementary exploratory data analysis figures

Appendix: EDA Summary

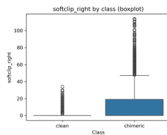
Table 4.1: Summary statistics of selected key features by class.

Feature	Class	Mean	Std	Median	IQR
has_sa	chimeric	0.406	0.491	0.0	1.0
has_sa	clean	0.000	0.000	0.0	0.0
num_segments	chimeric	1.406	0.491	1.0	1.0
num_segments	clean	1.000	0.000	1.0	0.0
softclip_left	chimeric	12.55	21.90	0.0	19.0
softclip_left	clean	0.23	1.54	0.0	0.0
softclip_right	chimeric	12.90	22.12	0.0	19.0
softclip_right	clean	0.21	1.51	0.0	0.0
total_clipped_bases	chimeric	25.44	25.48	19.0	48.0
total_clipped_bases	clean	0.44	2.16	0.0	0.0
kmer_js_divergence	chimeric	0.974	0.025	0.986	0.043
kmer_js_divergence	clean	0.976	0.025	0.986	0.040
kmer_cosine_diff	chimeric	0.974	0.026	0.986	0.042
kmer_cosine_diff	clean	0.976	0.025	0.986	0.041
microhomology_length	chimeric	0.458	0.755	0.0	1.0
microhomology_length	clean	0.462	0.758	0.0	1.0
microhomology_gc	chimeric	0.172	0.361	0.0	0.0
microhomology_gc	clean	0.172	0.361	0.0	0.0

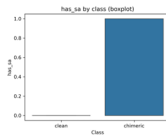
Appendix: Feature Boxplots



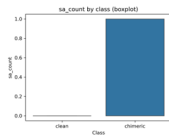
(a) Softclip left



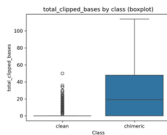
(b) Softclip right



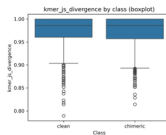
(c) Has SA



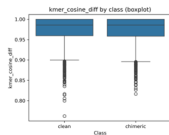
(d) SA count



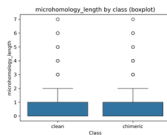
(e) Total clipped bases



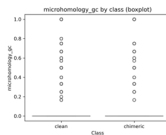
(f) K-mer JS divergence



(g) K-mer cosine difference



(h) Microhomology length



(i) Microhomology GC

Appendix: External Assembly Graphs (3,200 reads)

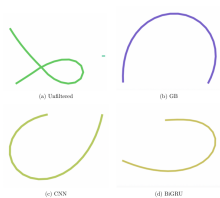


Figure F.1: Assembly graphs for 3200 paired-end reads at 5% chimera proportion.

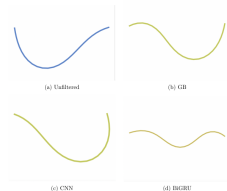


Figure F.2: Assembly graphs for 3200 paired-end reads at 10% chimera proportion.

5% chimeras

10% chimeras

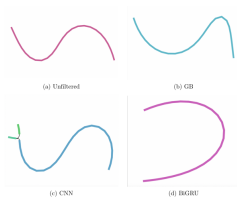


Figure F.3: Assembly graphs for 3200 paired-end reads at 15% chimera proportion.

15% chimeras

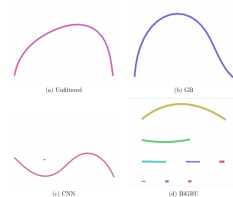


Figure F.4: Assembly graphs for 3200 paired-end reads at 50% chimera proportion.

50% chimeras

Appendix: External Assembly Graphs (10,000 reads)

FIG F.2.1 5% Chimera Proportion

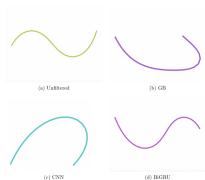


Figure F.2: Assembly graphs for 10K paired-end reads at 5% chimera proportion.

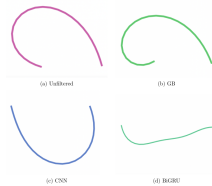


Figure F.3: Assembly graphs for 10K paired-end reads at 10% chimera proportion.

5% chimeras

10% chimeras

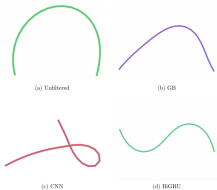


Figure F.7: Assembly graphs for 10K paired-end reads at 15% chimera proportion.

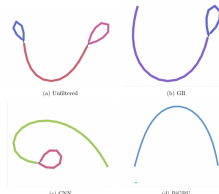


Figure F.8: Assembly graphs for 10K paired-end reads at 50% chimera proportion.

15% chimeras

50% chimeras

Appendix: External Assembly Graphs (20,000 reads)

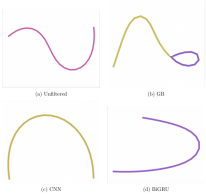


Figure F.9: Assembly graphs for 20K paired-end reads at 5% chimera proportion.

5% chimeras

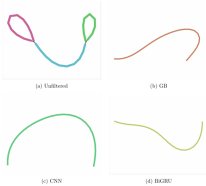


Figure F.10: Assembly graphs for 20K paired-end reads at 10% chimera proportion.

10% chimeras

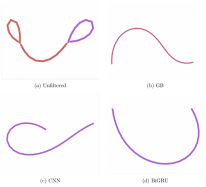


Figure F.11: Assembly graphs for 20K paired-end reads at 15% chimera proportion.

15% chimeras

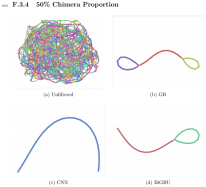


Figure F.12: Assembly graphs for 20K paired-end reads at 50% chimera proportion.

50% chimeras